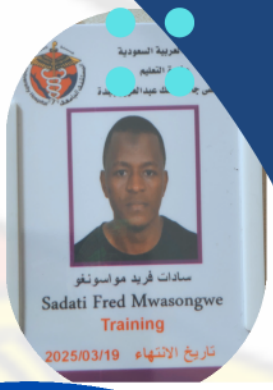
KING ABDULAZIZ UNIVERSITY
Faculty of Engineering

KAUH

## KING ABDULAZIZ UNIVERSITY HOSPITAL




# 2025 SUMMER TRAINING REPORT

/////// Biomedical Engineering ///////







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| START | END |
|---|---|
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# Abstract

This paper provides a complete description of my two-month summer biomedical engineering training program, which was held at King Abdulaziz University Hospital (KAUH) as part of King Abdulaziz University's (KAU) graduation requirements. The goal of this program was to obtain practical experience in clinical settings and to comprehend the critical role that biomedical engineers play in maintaining, testing, and guaranteeing the safety of medical equipment across many hospital departments. Throughout the training, I was able to tour and witness operations in several critical departments, including Radiology, Laboratory, Central Sterile Services Department (CSSD), Intensive Care Units (ICU), Dialysis, Operating Room (OR), and Outpatient Clinics. In each department, I explored a diverse spectrum of medical gadgets, including MRI machines, CT scanners, autoclaves, dialysis machines, anesthetic workstations, and ophthalmic diagnostic tools. Preventive and corrective maintenance, equipment calibration, infection control methods, and the value of teamwork across biomedical teams and clinical staff were all emphasized. Although direct hands-on involvement was occasionally limited due to equipment warranties and clinical constraints, I obtained useful insights through observation, technical talks, and documentation examination. This program really improved my technical understanding, career development, and respect for the critical role of biomedical engineers in ensuring safe and effective healthcare delivery.

# INTRODUCTION

**Overview of King Abdulaziz University Hospital (KAUH) as a Teaching Hospital**

King Abdulaziz University Hospital (KAUH) is one of the leading medical institutions in Saudi Arabia, known for its role as both a healthcare provider and an academic training center. Affiliated with King Abdulaziz University, the hospital plays a crucial role in medical education, research, and advanced patient care. As a teaching hospital, KAUH provides hands-on clinical training to medical students, nursing staff, and biomedical engineering trainees, ensuring that they gain real-world experience in a dynamic healthcare environment. The hospital houses a wide range of specialized departments, including Intensive Care Units (ICUs), Operating Rooms (ORs), Dialysis, Radiology, and Laboratory services, among others. With state-of-the-art medical equipment and a strong focus on technological advancements, KAUH not only delivers high-quality patient care but also fosters innovation in healthcare practices.

**Purpose and Objectives of the Training**

The primary purpose of my training at KAUH was to successfully complete the mandatory Summer Training course, a core requirement for graduating from my Biomedical Engineering program. This internship was essential in bridging the gap between academic knowledge and practical application, providing me with hands-on experience in a real hospital environment.

Beyond fulfilling the graduation requirement, this training aimed to:

- Enhance my technical skills in biomedical engineering by working with medical devices, maintenance systems, and hospital workflows.
- Develop my problem-solving abilities in troubleshooting, repairing, and calibrating medical equipment used in various hospital departments.
- Familiarize myself with hospital safety protocols, including fire safety, radiation safety, infection control, and emergency procedures.
- Understand the biomedical engineering workflow, including corrective and preventive maintenance (CM & PPM), and the acceptance testing of new medical devices.
- Improve my communication and teamwork skills by collaborating with healthcare professionals, biomedical engineers, and technical staff.

- Gain exposure to multiple hospital departments, such as ICU, Dialysis, Radiology, CSSD, and OPD, to develop a holistic understanding of biomedical engineering applications in healthcare.

This training was an invaluable opportunity to apply the theoretical knowledge acquired in my academic courses to real-world scenarios, preparing me for my future career as a biomedical engineer.

**Importance of Biomedical Engineering in a Hospital Setting**

Biomedical engineering is a critical component of modern healthcare, ensuring that medical equipment operates safely, efficiently, and reliably. Hospitals rely on advanced medical technologies for diagnosis, treatment, patient monitoring, and surgical procedures, and biomedical engineers play a pivotal role in managing these technologies.

The responsibilities of biomedical engineers in a hospital setting include:

- Medical Equipment Maintenance & Safety: Ensuring that ventilators, dialysis machines, infusion pumps, ECG monitors, and other critical devices function correctly and meet safety standards.
- Installation & Commissioning of Medical Devices: Setting up new medical equipment, performing initial testing, and ensuring proper integration into the hospital's workflow.
- Calibration & Performance Verification: Regularly calibrating devices such as patient monitors, infusion pumps, defibrillators, and anesthesia machines to ensure accuracy and reliability in medical procedures.
- Troubleshooting & Repair: Diagnosing technical failures, resolving errors, and repairing faulty medical equipment to minimize downtime and ensure patient safety.
- Technology Integration: Assisting in the implementation of new medical technologies, electronic health records, and hospital automation systems.
- Regulatory Compliance & Quality Control: Ensuring that all hospital equipment adheres to international safety standards such as ISO, IEC, and FDA regulations.
- Medical Equipment Procurement & Testing: Evaluating, testing, and approving new devices before they are deployed in the hospital.
- Supporting Clinical Staff: Collaborating with doctors, nurses, and technicians to provide technical assistance and training on medical equipment.

In essence, biomedical engineers serve as the bridge between medical technology and patient care, contributing significantly to improving healthcare delivery, reducing equipment downtime, and ensuring patient safety. My training at KAUH allowed me to experience these responsibilities firsthand, reinforcing the importance of my role in the healthcare system.

# TRAINING PLAN & HOSPITAL TOUR

**New Employee Orientation & Hospital Tour**

At the beginning of my summer training at King Abdulaziz University Hospital (KAUH), I participated in the new employee orientation program, which provided an overview of the hospital's structure, policies, and safety procedures. The orientation was essential in familiarizing me with the hospital environment, workflows, and regulations governing biomedical engineering operations.

As part of the hospital tour, I was introduced to the various medical departments and facilities, including:

- Intensive Care Units (ICUs) – Housing critically ill patients and equipped with ventilators, patient monitors, infusion pumps, and other life-supporting devices.
- Emergency Room (ER) – Handling acute medical cases with a range of diagnostic and life-saving equipment.
- Operating Rooms (ORs) – Specialized surgical areas where I observed high-tech anesthesia machines, electrosurgical units, and endoscopy systems.
- Dialysis Unit – Dedicated to patients undergoing renal replacement therapy, including hemodialysis machines and reverse osmosis (RO) water treatment systems.
- Radiology Department – Containing advanced imaging modalities such as X-ray, CT scan, MRI, ultrasound, and linear accelerators.
- Central Sterile Services Department (CSSD) – Responsible for the sterilization and disinfection of surgical instruments and hospital tools.
- Clinical Laboratories – Performing diagnostic tests in microbiology, hematology, virology, and immunology, utilizing automated analyzers and sample testing machines.
- Outpatient Clinics (OPD) – Providing routine medical consultations, including specialized clinics such as ENT, ophthalmology, OB-GYN, and cardiology.

This tour gave me valuable insights into how biomedical engineering integrates into hospital operations and helped me understand how different medical devices function within each department.

**Biomedical Engineering Department Training Plan**

During my two-month training period, I was assigned to the Biomedical Engineering Department (BME), where I engaged in both theoretical learning and practical hands-on experience. The training followed a structured plan covering multiple aspects of biomedical engineering, including medical equipment maintenance, safety protocols, troubleshooting techniques, and regulatory compliance.

The key training topics included:

**1. Biomedical Equipment Management**

- Understanding the hospital's Operation & Maintenance System (OMS) for tracking medical device history, maintenance schedules, and failure reports.
- Learning about the Corrective Maintenance (CM) and Planned Preventive Maintenance (PPM) programs, which ensure optimal performance of hospital equipment.
- Familiarizing myself with the acceptance and commissioning process for new medical equipment, including performance verification, calibration, and safety testing before deployment.

**2. Medical Equipment Maintenance & Testing**

I had hands-on experience working with various medical devices, performing:

- Installation & Setup: Assisting in the deployment of new medical devices, including patient monitors, defibrillators, and ventilators.
- Calibration & Testing: Using biomedical testing equipment such as ECG simulators, defibrillator testers, and infusion pump analyzers to ensure accurate readings and proper device functionality.
- Troubleshooting & Repairs: Diagnosing faulty sensors, display malfunctions, and power supply issues in patient monitoring devices and infusion pumps.
- Preventive Maintenance Procedures: Conducting routine cleaning, battery checks, sensor replacements, and firmware updates for critical hospital equipment.

### 3. Safety & Compliance in Biomedical Engineering

Patient safety and regulatory compliance are critical aspects of biomedical engineering in a hospital setting. King Abdulaziz University Hospital (KAUH) adheres to both national and international accreditation standards, ensuring that all medical equipment and hospital procedures comply with strict safety regulations.

Accreditations & Regulatory Standards

KAUH is accredited both nationally and internationally, demonstrating its commitment to high-quality healthcare and patient safety:

- CBAHI (Saudi Central Board for Accreditation of Healthcare Institutions) – This is the national accreditation for healthcare facilities in Saudi Arabia, ensuring that hospitals meet rigorous safety and quality standards. Biomedical engineers play a vital role in ensuring compliance with CBAHI standards, particularly in medical equipment management, maintenance protocols, and safety inspections.
- Accreditation Canada (CANADA Accreditation) – This international accreditation ensures that KAUH follows global best practices in healthcare quality, patient safety, and biomedical engineering processes. It requires continuous evaluation of medical devices, risk management strategies, and adherence to international safety protocols.

**Biomedical Engineering Safety Practices**

To comply with these standards, my training at KAUH focused on the following safety and regulatory aspects:

- Fire & Electrical Safety – Understanding emergency procedures, fire hazards in medical equipment, and proper grounding techniques to prevent electrical accidents.
- Radiation Protection – Learning safety measures in X-ray, CT, and MRI departments, including shielding techniques and personal protective equipment (PPE) to minimize radiation exposure.
- Medical Equipment Risk Assessment – Conducting regular safety audits to identify potential hazards in ICU, ER, and OR devices, ensuring that all equipment is functioning within regulatory limits.
- Preventive Maintenance & Calibration – Ensuring defibrillators, ventilators, patient monitors, and infusion pumps are regularly tested using biomedical analyzers to maintain compliance with CBAHI and CANADA Accreditation guidelines.

By working in multiple hospital departments, I witnessed how biomedical engineering contributes to maintaining a safe healthcare environment, ensuring that all medical devices comply with national and international accreditation standards.

**4. Specialized Training in Key Hospital Departments**

- Dialysis Unit: Observed the operation of dialysis machines and RO water treatment systems, ensuring proper fluid filtration and patient safety.
- ICU & ER Equipment: Assisted in checking ventilators, patient monitors, and infusion pumps, verifying correct readings and alarm functions.
- Operating Rooms (ORs): Learned about electrosurgical units, anesthesia machines, and endoscopy towers, ensuring proper calibration and safe operation.
- Radiology Department: Observed MRI, CT, and X-ray machine maintenance, including troubleshooting software and hardware issues.
- Clinical Laboratories: Gained insights into automated diagnostic analyzers, blood gas analyzers, and centrifuges, ensuring precision in medical testing.

This structured training plan provided me with extensive hands-on experience, allowing me to develop a deep understanding of biomedical engineering operations in a hospital setting.

The hospital tour and training orientation were crucial in helping me navigate KAUH's medical environment, safety protocols, and biomedical engineering workflows. By being exposed to real-world medical equipment challenges, I was able to enhance my technical skills, problem-solving abilities, and overall readiness for a future career in biomedical engineering.

# DEPARTMENTAL TRAINING EXPERIENCE

During my training at King Abdulaziz University Hospital (KAUH), I had the opportunity to work across multiple critical hospital departments, each with its own specialized medical equipment and maintenance requirements. This section provides an in-depth explanation of my hands-on experience, key biomedical devices encountered, maintenance and troubleshooting procedures, and challenges faced in each department.

Each subsection will cover:

- Department Overview: The role and function of the department in patient care.
- Medical Equipment Used: Key devices I worked with and their significance.
- Maintenance & Troubleshooting Procedures: Steps taken to ensure proper functionality, including calibration, testing, and repairs.
- Challenges & Solutions: Issues encountered during equipment maintenance and how I resolved them.

| | | |
|---|---|---|
| **Biomedical Engineering** | **Emergency & ICU** | **Operating Rooms & Anesthesia** |
| **Radiology & Imaging** | **Dialysis & Nephrology** | **Oncology & Radiotherapy** |
| **Ophthalmology & ENT** | **Dental & Maxillofacial** | **Medical Rehabilitation** |
| **Surgery (General & Specialized)** | **Pharmacy & Laboratory** | **Obstetrics & Gynecology** |
| **Neonatal & Pediatrics** | **Sterilization & Infection Control** | **Home Healthcare & Patient Support** |
| **Medical Records & IT Services** | **Security & Safety** | **Outpatient Clinics** |

## Intensive Care Units (ICUs)

During my training at King Abdulaziz University Hospital (KAUH), I had the opportunity to work in the Intensive Care Units (ICUs), where critically ill patients receive advanced medical care. The ICU departments are equipped with highly specialized medical devices that require continuous monitoring, maintenance, and calibration to ensure patient safety and effective treatment.

KAUH has multiple ICU sections, each catering to different patient needs and age groups. The hospital follows a structured ICU layout across different floors, as outlined below:

- **Ground Floor:**
  - ❖ Surgical Intensive Care Unit (SICU): Specializes in post-operative care for critically ill surgical patients.
  - ❖ Coronary Care Unit (CCU): Focuses on patients with severe cardiac conditions, including heart attacks and arrhythmias.
- **1st Floor:**
  - ❖ Neonatal Intensive Care Unit (NICU): Dedicated to premature and critically ill newborns, requiring incubators, ventilators, and phototherapy units.
- **2nd Floor:**
  - ❖ Medical Intensive Care Unit (MICU): Provides care for patients with severe medical conditions such as respiratory failure, sepsis, and organ dysfunction.
  - ❖ Pediatric Intensive Care Unit (PICU): Treats critically ill children requiring advanced life support.
  - ❖ Pediatric Cardiac Intensive Care Unit (PCICU): Specializes in pediatric patients with congenital and acquired heart diseases.
- **3rd Floor:**
  - ❖ Pediatric Intensive Care Unit II (PICU II): Serves as a backup ICU when PICU on the 2nd floor reaches full capacity.

## Medical Equipment Used in ICUs

During my rotation in the ICUs, I worked with and observed the maintenance, calibration, and troubleshooting of several critical medical devices, including:

| Device | Image | Function | My responsibility |
|---|---|---|---|
| **Ventilators** | | Provides mechanical breathing support to patients who cannot breathe independently. | Checked alarm functions, airflow calibration, and oxygen concentration testing. |
| **Patient Monitors** | | Monitors vital signs such as heart rate, blood pressure, oxygen saturation, and temperature. | Ensured accuracy of readings through **calibration** and sensor testing. |
| **Infusion & Syringe Pumps** | | Administers fluids, medications, or nutrients into a patient's bloodstream in controlled amounts. | Tested **flow rate accuracy** and checked for motor malfunctions. |
| **Pulse Oximeters** | | Measures blood oxygen levels and pulse rate. | Verified **$SpO_2$ sensor accuracy** using a biomedical simulator. |
| **ECG Monitors** | | Records the electrical activity of the heart. | Troubleshot **lead connectivity issues** and tested waveform accuracy. |
| **Defibrillators** | | Delivers controlled electric shocks to restore normal heart rhythm. | Performed **energy output verification** using a defibrillator analyzer. |
| **Blood Pressure Monitors** | | Measures systolic and diastolic blood pressure. | Checked **cuff inflation** and **pressure sensor calibration**. |
| **Incubators (NICU)** | | Provides temperature-controlled environments for premature infants. | Ensured **temperature stability** and **alarm system functionality**. |
| **Cardiac Output Monitors (CCU)** | | Measures the amount of blood the heart pumps through the circulatory system. | Checked **sensor accuracy** and **catheter function verification**. |
| **Mechanical Suction Units (SICU)** | | Removes secretions, blood, or other fluids from the airway or surgical site. | Verified **vacuum pressure** and tested **canister seal integrity**. |
| **High-Frequency Ventilators (NICU/PICU)** | | Offers specialized ventilation for neonates and pediatric patients with delicate lungs. | Calibrated **oscillator settings** and tested **circuit pressure limits**. |

| | | | |
|---|---|---|---|
| **Intra-Aortic Balloon Pumps (CCU/PCICU)** | | Assists the heart in pumping more blood by inflating and deflating a balloon in the aorta. | Verified **timing synchronization** and performed **safety checks**. |
| **Blood Gas Analyzers (MICU)** | | Analyzes levels of oxygen, carbon dioxide, and pH in blood samples. | Performed **reagent calibration**, quality control, and **sample probe cleaning**. |
| **Capnography Monitors (MICU/SICU)** | | Measures the concentration of carbon dioxide in exhaled air. | Tested **end-tidal $CO_2$ accuracy** and calibrated infrared sensors. |
| **Phototherapy Units (NICU)** | | Treats neonatal jaundice by using light to break down bilirubin in the skin. | Checked **light intensity**, spectral range, and **timer accuracy**. |
| **External Pacemakers (PCICU)** | | Provides electrical pacing to patients with arrhythmias or heart blocks. | Verified **pacing threshold**, lead output, and **battery integrity**. |
| **Oxygen Concentrators (All ICUs)** | | Supplies concentrated oxygen to patients. | Checked **oxygen purity**, **alarm response**, and **flow regulator function**. |
| **Heated Humidifiers (PICU/NICU)** | | Adds moisture and warmth to gases delivered through ventilators. | Calibrated **temperature sensors** and tested **humidity delivery systems**. |



*Figure 1 An ICU unit room*

## Maintenance & Testing Procedures

- Preventive Maintenance (PPM): Conducted routine maintenance on ventilators, monitors, and infusion pumps to prevent unexpected failures.
- Calibration & Accuracy Checks: Used test equipment (such as ECG simulators and infusion pump analyzers) to ensure devices provided accurate readings.
- Troubleshooting & Repairs: Diagnosed and resolved sensor failures, power supply issues, and alarm malfunctions.

## Challenges Faced & How I Overcame Them

During my training in the Intensive Care Units (ICUs), I encountered several challenges related to equipment maintenance and access restrictions:

- Manufacturer Warranty Restrictions: Most ICU medical devices are covered under manufacturer warranties, meaning I was not allowed to perform maintenance, repairs, or internal modifications. Only certified engineers from the respective companies were authorized to troubleshoot and fix the devices.
- Limited Hands-On Experience: In some cases, when a device malfunctioned, the company's engineer either fixed it on-site while I observed or removed the device for external servicing, preventing me from gaining direct repair experience.
- Troubleshooting Without Physical Access: Since I couldn't open or repair certain devices, my role was often limited to identifying issues, checking error logs, and reporting problems to the engineers.

**How I Overcame These Challenges**

- Active Observation: Whenever an external engineer performed repairs, I closely observed the diagnostic procedures, tools used, and software interfaces, asking relevant questions to understand the process.
- Testing & Calibration Tasks: While I couldn't repair devices under warranty, I was allowed to perform functional tests, calibrations, and preventive maintenance checks, ensuring proper operation before returning the devices to clinical use.
- Learning from Documentation: I studied the user manuals, service logs, and troubleshooting guides provided for each device to enhance my theoretical understanding of common failures and repair techniques.

Despite these limitations, I gained valuable insights into ICU equipment management, learning how biomedical engineers coordinate with medical equipment manufacturers to ensure optimal device performance and patient safety.

| Floor | ICU Section | Specialization |
|---|---|---|
| Ground Floor | SICU (Surgical ICU) | Post-surgical critical care. |
| | CCU (Coronary Care Unit) | Cardiac emergencies and heart failure cases. |
| 1st Floor | NICU (Neonatal ICU) | Premature and critically ill newborn care. |
| 2nd Floor | MICU (Medical ICU) | Critical care for medical conditions (respiratory failure, sepsis, etc.). |
| | PICU (Pediatric ICU) | Intensive care for children. |
| | PCICU (Pediatric Cardiac ICU) | specialized cardiac ICU for pediatric heart conditions. |
| 3rd Floor | PICU II (Backup Pediatric ICU) | Overflow ICU for pediatric patients. |

## Radiology

I had the opportunity to gain hands-on experience in the Radiology Department. This department is a critical component of patient diagnosis and treatment planning, encompassing several high-tech imaging modalities. As a biomedical engineering trainee, I worked alongside clinical engineers and radiology technologists to ensure that all imaging equipment operated safely and efficiently.



Medical Equipment Used in Radiology (Organized by Modality)

| Modality | Key Medical Equipment | Purpose |
|---|---|---|
| **General X-Ray & Fluoroscopy** | - Digital X-ray machine | - Static imaging of bones and lungs |
| | - Fluoroscopy C-arm system | - Real-time imaging (fluoroscopy) |
| | - Wall bucky | |
| | - Mobile X-ray unit | |
| **Digital Mammography** | - Full-field digital mammography unit | - Early detection of breast cancer |
| | - Compression paddles | |
| **C.T. Scan** | - Multi-slice CT scanner | - Cross-sectional imaging for trauma, tumors, and vascular diseases |
| | - Contrast injector system | |
| **Ultrasound** | - Portable ultrasound unit | - Real-time imaging for OB/GYN, abdominal, cardiac, and vascular exams |
| | - Doppler ultrasound | |
| | - Transducers | |
| **MRI** | - 1.5T / 3T MRI system | - Detailed soft tissue imaging using magnetic fields and radio waves |
| | - Gradient coils | |
| | - RF coils | |
| | - Cooling system | |
| **Gamma Camera** | - SPECT gamma camera | - Nuclear medicine imaging for thyroid, bones, and cardiac studies |
| | - Radiopharmaceutical injectors | |
| **Linear Accelerator** | - LINAC machine | - High-energy radiation for cancer treatment (External Beam Radiation) |
| | - Multileaf collimator (MLC) | |
| | - Patient positioning system | |
| **Brachytherapy System** | - HDR afterloader | - Internal radiation therapy, mainly for cervical, prostate, and breast cancers |
| | - Treatment planning system | |
| | - Applicators | |

## Maintenance & Testing Procedures

As a biomedical engineering trainee, I actively participated in Preventive Maintenance (PM) and Corrective Maintenance (CM) tasks, as well as acceptance testing and performance verification procedures. Here's a breakdown:

| Procedure | Description |
|---|---|
| **Preventive Maintenance (PM)** | Scheduled checkups for cooling systems, oil levels (e.g., X-ray tubes), mechanical parts, and filters |
| **Electrical Safety Testing** | Used multimeter and safety analyzers to verify grounding, leakage currents, and insulation |
| **Image Quality Testing** | Used phantoms (e.g., CT, MRI, ultrasound) to check contrast resolution, spatial resolution, and artifacts |
| **Calibration** | Adjusted parameters like tube current, exposure time, gantry alignment, and CT Hounsfield Units |
| **Software Updates** | Performed under supervision for MRI/CT consoles and PACS interface modules |
| **Magnet Field Check (MRI)** | Verified magnetic field homogeneity and temperature-controlled cooling (cryogen system) |
| **Gamma Camera QC** | Daily uniformity check, weekly bar phantom test for resolution |
| **Radiation Leakage Testing** | Used radiation survey meters to ensure shielding and exposure levels were within limits |

## Challenges Faced & How I Overcame Them

| Challenge | How I Overcame It |
|---|---|
| **Understanding complex imaging systems (MRI & CT)** | I requested detailed walkthroughs from senior engineers and reviewed equipment manuals and training videos |
| **Radiation safety protocol unfamiliarity** | I studied IAEA and ICRP safety standards, shadowed radiology technologists, and used dosimeters properly |
| **Limited access to clinical rooms (infection control / radiation exposure)** | I coordinated with department supervisors and scheduled maintenance during off-peak hours |
| **Software troubleshooting (DICOM & PACS issues)** | I learned basic network mapping and got involved in PACS system restarts and testing |
| **Lack of hands-on with Brachytherapy system** | I observed treatment planning sessions and reviewed QA protocols with physicists |
| **Cryogen handling in MRI (helium refills / quenching concerns)** | I attended a safety orientation and avoided entering Zone 4 unsupervised |

During my training in the Radiology Department at King Abdulaziz University Hospital, I learned about a variety of imaging technologies, including portable ultrasound systems, CT scanners, MRI machines, and, most notably, Linear Accelerator (LINAC), the largest and most expensive device used in cancer treatment. I learned that radiology is divided into two categories: diagnostic radiology, which focuses on disease detection using modalities such as X-ray, CT, MRI, ultrasound, and nuclear medicine, and therapeutic radiology, which includes treatment tools such as LINAC and brachytherapy systems. While both branches rely on improved imaging and precision, their primary tasks differ greatly. I also recognized the crucial need of safety in this profession, particularly radiation protection, and implemented techniques such as shielding, equipment calibration, radiation leakage testing, and adherence to correct safety regulations. This experience expanded my technical knowledge and strengthened my awareness for the critical role that biomedical engineers play in upholding high standards of care and safety in medical imaging settings.

## Laboratory department

Despite the fact that I did not have much hands-on experience during my rotation in the Laboratory Department at King Abdulaziz University Hospital, I was able to obtain a general understanding of the department's workflow and the several specialist departments it houses. My visits included microbiology, blood bank, virology, immunology, biochemistry, hematology, and histopathology. Each division had a specific purpose and used a range of diagnostic analyzers, centrifuges, incubators, and microscopes, among other necessary equipment. I noticed the organizational structure and how biomedical equipment is essential for accurately and rapidly processing and evaluating patient samples. This experience taught me about the complexities and importance of laboratory work in supporting proper diagnosis and treatment plans. Despite the limited engagement, this experience helped me understand the many instruments utilized, their functions in clinical diagnostics, and the importance of frequent maintenance and calibration in assuring accurate lab results.



## Medical Equipment Used in Laboratory

| Laboratory Section | Medical Equipment Observed / Commonly Used |
|---|---|
| **Microbiology** | Incubators, Biological safety cabinets (BSC), Autoclaves, Streaking robots, Microscopes, Colony counters, Anaerobic jars, Petri dish dispensers |
| **Blood Bank** | Blood centrifuges, Blood refrigerators, Plasma freezers, Blood bank analyzers, Gel card centrifuges, Blood agitators, Tube sealers, Cross-matching workstations |
| **Virology** | Real-Time PCR machines, Biosafety cabinets, Refrigerated centrifuges, Viral transport systems, ELISA readers, Microplate washers, $CO_2$ incubators |
| **Immunology** | Immunoassay analyzers (e.g., Architect, Cobas), ELISA systems, Centrifuges, Incubators, Sample preparation systems |
| **Biochemistry** | Fully automated chemistry analyzers, Sample/reagent racks, pH meters, Electrolyte analyzers, Spectrophotometers, Water baths, Refrigerated centrifuges |
| **Hematology** | Hematology analyzers (CBC machines), ESR analyzers, Cell counters, Blood smearing devices, Slide stainers, Microscopes |
| **Histopathology** | Microtomes, Tissue processors, Embedding stations, Slide stainers, Cover slippers, Cryostats, Paraffin dispensers, Light microscopes, Fume hoods |

Although my experience in the Laboratory Department at King Abdulaziz University Hospital was more observational than hands-on, it gave me with a broad grasp of the many diagnostic technologies used in different divisions. From simple instruments like centrifuges and microscopes to modern analyzers and automation systems, I saw how biomedical equipment improves the accuracy and efficiency of laboratory testing. Each section, whether Microbiology, Blood Bank, Virology, Immunology, Biochemistry, Hematology, or Histopathology, depends largely on well-maintained technology to ensure accurate results. This experience increased my understanding of laboratory operations and the critical role that biomedical engineers play in maintaining and assuring the functionality of diagnostic equipment.

## Central Sterile Services Department (CSSD)

During my training at KAUH's Central Sterile Services Department (CSSD), I observed the whole cycle of medical instrument reprocessing, from receiving contaminated tools after use to cleaning, sterilization, packaging, and redistribution to various hospital departments. The CSSD is a highly controlled environment that is critical for infection protection. Biomedical engineers play an important role in ensuring that all surgical and medical instruments are thoroughly cleaned, sanitized, and ready for reuse.

### Main Functional Areas & Equipment

| Area | Equipment | Function |
|---|---|---|
| **Decontamination Area** | **Washer Disinfector** | Automatically washes and disinfects surgical tools using thermal and chemical cycles. |
| | **Ultrasonic Cleaner** | Uses ultrasonic waves and enzymatic detergents to remove microscopic debris from complex instruments. |
| | **Cart Washer** | Cleans large surgical carts, containers, and trays using high-pressure water and detergent. |
| **Preparation & Packing Area** | - Workstations with instrument trays, inspection lights, sealing devices | Technicians inspect, assemble, and package cleaned instruments. |
| **Sterilization Area** | **Autoclave (Steam Sterilizer)** | Uses high-pressure saturated steam (121–134°C) to sterilize heat-resistant tools. |
| | **Plasma Sterilizer (Low Temp)** | Uses hydrogen peroxide gas plasma for heat- and moisture-sensitive devices. |
| | **Ethylene Oxide (EtO) Sterilizer** *(optional, if available)* | For sensitive instruments not suitable for heat or moisture; very effective but toxic and time-consuming. |
| | **Steam Generator** | Supplies consistent, high-quality steam to the autoclaves. |

## Training Experience

As a trainee in biomedical engineering, I paid great attention to the decontamination process and became familiar with each device's technical details, cycles of operation, and control interfaces. Strict safety procedures prevented me from physically operating the machines, but I did learn how biomedical personnel test, check, and repair devices. I discovered how essential every piece of equipment is to guarantee sterile, infection-free tools for patient care, especially in intensive care units and surgical settings.

## Maintenance and Testing Insight

- Daily maintenance: Filter cleaning, cycle validation, water level checks.
- Weekly/monthly: Biological indicator tests, chamber pressure and temperature calibration.
- Biomedical role: Ensuring pressure sensors, temperature sensors, door locks, and steam generators function safely and efficiently.

## Summary

I learned a lot about the importance of sterilization in infection control during my training in the CSSD department at KAUH. I saw how several devices, including plasma sterilizers, autoclaves, ultrasonic cleaners, and washer disinfectors, cooperate to guarantee that all medical equipment is safe for reusing. Strict sterilizing procedures prevented me from having much hands-on experience, but I did have a thorough understanding of the equipment's operation, workflow, and the significance of regular testing and maintenance. This event made clear how crucial biomedical engineers are to preserving sterile services' effectiveness, functioning, and safety.

## Outpatient Department (OPD)

During my training at the OPD at KAUH, I had the opportunity to observe the operation of numerous specialist outpatient clinics, including Medical, Surgical, Eye, OB-Gyne, and ENT clinics. These departments are the primary point of contact for patients who do not require emergency care or hospitalization. I learnt about the various biomedical instruments used in diagnostics and minor procedures, as well as the critical role that equipment readiness and maintenance play in maintaining fast patient flow and correct diagnosis.

### Key OPD Clinics & Equipment Overview

| Clinic | Common Devices Observed | Function |
|---|---|---|
| **Medical Clinic** | Digital Thermometers, BP Monitors, Pulse Oximeters, ECG Machines, Spirometers | Basic diagnostics for general medical cases |
| **Surgical Clinic** | Minor Procedure Lights, Electrosurgical Units, Suction Devices, Sterile Instrument Sets | For wound care, minor surgeries, and follow-ups |
| **Eye Clinic** | Slit Lamp, Autorefractor, Tonometer, OCT (Optical Coherence Tomography), Visual Field Analyzer, Lensometer, Phoropter | Detailed ophthalmic examination, diagnosis of glaucoma, retinal imaging, and vision testing |
| **OB-GYNE Clinic** | Fetal Doppler, Portable Ultrasound, Examination Tables with stirrups, Colposcope | Monitoring pregnancy, pelvic exams, and minor gynecological procedures |
| **ENT Clinic** | Endoscopes (Rigid & Flexible), Tympanometers, Audiometers, Nasal Suction, Otoscope | Examining and diagnosing ear, nose, and throat conditions |

## Special Focus – Eye Clinic Training

The Eye Clinic provided me with a deeper and more hands-on learning experience compared to the other clinics. I closely observed and learned the operation of advanced ophthalmology equipment such as:

- Slit Lamp – Used for examining the anterior segment of the eye in detail.
- Autorefractor – Automatically measures refractive error for prescription glasses.
- Tonometer – Measures intraocular pressure, essential in diagnosing glaucoma.
- OCT – Non-invasive imaging that provides cross-sectional views of the retina.
- Visual Field Analyzer – Assesses peripheral vision; crucial in glaucoma management.
- Phoropter and Lensmeter – For checking and prescribing corrective lenses.

I gained valuable insights into the calibration procedures, daily checks, and cleaning requirements for each device. Understanding how even small alignment issues can impact diagnostic accuracy reinforces the importance of biomedical support.

## Maintenance & Testing Procedures

- Routine checks for calibration (especially in vision and pressure-based devices).
- Cleaning and disinfection protocols, especially for reusable contact parts.
- Light source inspections (e.g., in slit lamps, colposcopes, endoscopes).
- Battery and cable inspections in portable or handheld devices.
- Biomedical engineers also ensure software updates, and firmware calibration when necessary.

## Challenges Faced

- Limited space in some clinics makes movement and servicing of equipment a bit difficult.
- High patient volume often limits the time available for thorough device checks during the day.
- I overcame this by assisting during early or off-peak hours and observing device usage closely to better understand wear-and-tear patterns.

During my training in the Outpatient Department (OPD) at KAUH, I had the opportunity to explore various specialized clinics including Medical, Surgical, OB-Gyne, ENT, and Eye Clinics. Although my interaction with some clinics was brief, I gained valuable insights into the diversity of diagnostic and treatment devices used in outpatient care. My most in-depth experience was in the Eye Clinic, where I observed and learned about high-precision ophthalmic equipment like slit lamp, autorefractor, OCT, and tonometer. This exposure highlighted the importance of regular calibration, hygiene protocols, and software checks to ensure diagnostic accuracy and patient safety. Overall, the OPD experience deepened my understanding of the vital role biomedical engineers play in maintaining the reliability and efficiency of equipment in high-turnover clinical environments.



## Operating Room (OR) Department

My training in the **Operating Room (OR)** department at KAUH allowed me to observe a highly advanced surgical environment that integrates cutting-edge technologies with strict sterilization and workflow protocols. The OR is one of the most equipment-intensive departments in any hospital, and I witnessed how crucial it is for all systems to function seamlessly to ensure patient safety and successful surgical outcomes.



## OR Sections & Equipment Overview

| Category | Equipment/Systems | Function / Role |
|---|---|---|
| **Digital OR** | Integrated Operating Room System, Wall-mounted Display Monitors, Video Routing System | Streamlines communication, documentation, and real-time display of surgical field |
| **Electrosurgery Units** | Monopolar & Bipolar Electrosurgical Units (ESUs), Smoke Evacuators | Used for cutting and coagulation of tissue using high-frequency current |
| **Surgical Instruments** | General Surgery Sets, Orthopedic Tools, Neurosurgery Kits, Sterile Drapes & Trays | Instruments specific to the type of surgery being performed |
| **Anesthesia Equipment** | Anesthesia Workstations, Ventilators, Syringe & Infusion Pumps, Gas Scavenging Systems | For delivering and monitoring anesthesia gases and patient vitals |
| **Operating Microscope** | Surgical Microscopes (ENT, Neurosurgery, Ophthalmic), with Zoom and Light Source | Allow enhanced visualization of delicate anatomical structures |
| **Endoscopy Towers** | Laparoscopic Towers, Light Source, Insufflator, Camera Unit, Monitors | For minimally invasive surgeries such as laparoscopy or arthroscopy |
| **Laser Systems** | $CO_2$ Laser, Nd:YAG Laser, Argon Laser | Used in precision cutting, ablation, or cauterization depending on the surgery type |

## Maintenance & Testing Procedures

| Task | Description |
|---|---|
| **Functional Testing** | Power-on checks, test cycles, and safety tests (especially ESUs and anesthesia machines) before every surgery day |
| **Calibration** | Regular calibration of anesthesia ventilators and laser systems as per manufacturer specs |
| **Sterility Assurance** | Instruments and endoscopy components are checked for proper sterilization tags and packing integrity |
| **Light Source & Display Check** | Ensure OR lights, surgical microscope lights, and monitors are working correctly |
| **Battery & Cable Inspections** | All portable devices like infusion pumps and scopes are checked for battery charge and cable damage |
| **Software Updates** | Digital OR systems and anesthesia machines are checked for firmware/software updates periodically |

## Challenges Faced & How I Overcame Them

During my training in the Operating Room (OR) department, I faced several challenges that limited direct hands-on experience with the equipment. Many of the machines were under warranty, requiring maintenance and repairs to be carried out exclusively by externally certified engineers, which restricted my involvement in actual service tasks. Additionally, gaining access to the OR was not easy due to its continuous usage of surgical procedures and the strict hygiene and safety protocols that had to be always followed. These limitations made it difficult to observe equipment in active use or interact closely with clinical staff during procedures. To overcome these challenges, I focused on observational learning by reviewing maintenance logs, equipment manuals, and discussing technical aspects with biomedical staff. This allowed me to still gain valuable knowledge about the equipment's functions, safety standards, and the workflow of biomedical support within the OR environment.

## Dialysis Department

During my training in the Dialysis Department at King Abdulaziz University Hospital (KAUH), I was introduced to the critical role biomedical engineering plays in supporting patients with kidney failure. The department is divided into two main areas: the Water Treatment Room and the Dialysis Treatment Area. Together, they form the backbone of hemodialysis therapy.

### Water Treatment Room (RO Units)

The Water Treatment Room is one of the most critical components of the dialysis unit. It ensures that all dialysis water is extremely pure and free of pollutants that could harm patients. Dialysis requires significant volumes of water—up to 120 liters per session per patient—so water cleanliness is critical for patient safety. Reverse Osmosis (RO) is the primary method used to cleanse water before it enters the dialysis equipment.

| Equipment | Function |
|---|---|
| **Reverse Osmosis (RO) Unit** | Removes dissolved salts, organic materials, bacteria, and endotoxins from water. |
| **Carbon Filters** | Remove chlorine and chloramines to protect the RO membrane and patients. |
| **Sand Filters** | Eliminate sediment and larger particles from incoming water. |
| **Water Softener** | Reduces water hardness (calcium and magnesium) to protect downstream equipment. |
| **UV Sterilizer** | Kills bacteria and viruses using ultraviolet light. |
| **Conductivity Meters** | Monitor water purity at various points in the system. |
| **Storage Tank** | Stores treated water for use during dialysis sessions. |
| **Distribution Loop with Heat Disinfection** | Carries clean water to dialysis machines and is disinfected regularly. |
| **Heat Exchanger** | Transfers heat to raise water temperature during **thermal disinfection** of the RO loop, ensuring removal of biofilm and bacteria without chemical use. |

## Relation Between RO and Dialysis:

RO water is utilized directly to make the dialysate, which then combines with a concentrate to form a fluid that flows through the dialyzer. This fluid facilitates the diffusion of waste materials from the patient's blood across the dialyzer membrane. Any pollutants in the water (bacteria, endotoxins, or heavy metals) can enter the patient's system and cause major consequences. As a result, the RO unit is constantly monitored to ensure patient safety.

## Dialysis Equipment

Inside the dialysis treatment area, I observed various life-support devices and supporting technologies:

| Equipment | Function |
|---|---|
| **Hemodialysis Machines (e.g., Fresenius, B. Braun)** | Perform blood filtration by removing waste, excess fluid, and toxins. |
| **Dialyzers (Artificial Kidneys)** | Act as filters in the hemodialysis machine to clean the blood. |
| **Blood Pumps** | Regulate blood flow during dialysis. |
| **Ultrafiltration Control Units** | Manage fluid removal rates for each patient. |
| **Patient Monitors** | Track blood pressure, heart rate, and temperature throughout the procedure. |
| **Weighing Scales** | Monitor patient weight before and after dialysis sessions. |
| **Infusion Pumps** | Deliver medications or fluids during treatment. |
| **Reclining Dialysis Chairs** | Provide comfort during long treatment sessions and include patient access features. |

## Maintenance & Testing Procedures

Although my hands-on role was limited, I learned that regular **preventive maintenance (PPM)** is essential for both dialysis machines and water treatment systems. PPM includes checking pump pressures, flow rates, calibration, alarm testing, and replacing filters and membranes. Water quality is tested frequently for conductivity, chlorine, endotoxin levels, and bacterial contamination to meet international standards (such as AAMI).

## Challenges Faced & How I Overcame Them

Access to specific maintenance activities was restricted due to the sensitive nature of dialysis treatment and the constant use of devices throughout the day. In many cases, machinery was currently in use, and technicians were concerned with guaranteeing continued therapy. Furthermore, maintenance on the RO systems was scheduled at off-peak hours, limiting my direct exposure. However, I overcome these constraints by witnessing routine checks, asking questions, studying service logs, and becoming familiar with the biomedical team's equipment manuals and standard operating procedures.

## Discussion

Training in the Dialysis Department gave me a clear understanding of how crucial water quality and machine performance are to patient safety. I gained insights into both the engineering and clinical aspects of dialysis care. Observing the sophisticated integration between water treatment systems and hemodialysis machines helped me appreciate the precision required in this life-sustaining therapy. Learning about the role of RO water in generating dialysate deepened my understanding of the process and reinforced how biomedical engineering safeguards patient health on a daily basis.

## Medical Gas Supply – Gas Cylinders and Central Systems

During my training at King Abdulaziz University Hospital, one of the most fascinating and hands-on experiences I had was working in the Medical Gas Supply Unit, specifically with gas cylinders and central heating systems. This department is critical to the hospital's operations because medical gases are required for a variety of clinical applications such as surgery, intensive care, anesthesia, respiratory therapy, neonatal care, and emergency medicine. Even a little stoppage in gas flow can have fatal implications, especially for patients on ventilators, undergoing surgery, or requiring oxygen therapy. Under the guidance of my trainer, I was able to witness and learn about the various types of medical gases used throughout the hospital, as well as how they are carefully kept and maintained, and transferred to different departments via a complicated central pipeline system. I got an understanding for the engineering systems that ensure the continuous and regulated flow of life-sustaining gases, as well as the vital roles that biomedical engineers play in ensuring the systems' safety, continuity, and performance.

### Types of Medical Gases Observed

The hospital utilizes a wide variety of gases for clinical, surgical, diagnostic, and laboratory applications. Below is a breakdown of the major gases I observed:

| Gas Type | Purpose / Application |
|---|---|
| **Oxygen ($O_2$)** | Essential for respiration therapy, anesthesia, ICU, OR, and emergency care. |
| **Nitrous Oxide ($N_2O$)** | Used as an anesthetic gas in surgical procedures. |
| **Entonox ($O_2$ + $N_2O$ mix)** | Used for labor analgesia and minor surgical procedures. |
| **Carbon Dioxide ($CO_2$)** | Used in endoscopy, laparoscopic surgeries, and cryotherapy. |
| **Breathing Air** | Filtered compressed air is used in ventilators and other respiratory equipment. |
| **Liquid Oxygen (LOX)** | Stored in bulk tanks for continuous oxygen supply to critical areas. |
| **Liquid Nitrogen ($LN_2$)** | Used for cryopreservation, dermatology, and lab sample storage. |
| **Helium (He)** | Used in MRI systems (supercooling magnets) and respiratory therapy. |
| **Argon** | Specialty gas used in laser surgery and plasma coagulation. |
| **Gas Mixtures** | Custom blends used in labs or calibration of gas-sensitive equipment. |

## Cylinder Sizes and Storage

KAUH, being a large-scale tertiary hospital, houses a full range of cylinder sizes, from portable units to massive bulk tanks:

| Cylinder Size | Common Label | Capacity | Usage |
|---|---|---|---|
| **Small** | D size | ~350 liters | Transport or mobile use |
| **Medium** | E, F size | 600–1300 liters | ICU/OR backup, minor departments |
| **Large** | K size | ~7000 liters | Fixed room supply or higher demand usage |
| **Bulk Tanks** | LOX / $LN_2$ Tanks | Thousands of liters | Continuous hospital-wide supply through pipelines |

I was able to inspect all these sizes firsthand and understand how each is connected to supply systems or used as standalone backup units. Also, I had the chance to physically inspect these cylinders, understand the color coding, labeling standards (both American and International), and how pressure levels are monitored.

## Gas Supply & Distribution System

The hospital uses a central pipeline distribution system to deliver medical gases directly from the storage tanks to individual wards, ICUs, ORs, and departments. This system includes:

- Manifold Rooms: Where cylinders are connected in series to maintain continuous supply.
- Pressure Regulators and Gauges: Ensure gas pressure remains stable and safe.
- Alarm Panels: Installed throughout the hospital to detect leaks, low pressure, or supply failure.
- Zone Valves: Allow isolation of specific areas during maintenance or emergencies.
- Backup Cylinders: Strategically placed across departments for emergency use.

### Training Experience and Hands-On Involvement

Under the guidance of my trainer, I was able to participate in:

- Inspecting gas cylinder labeling, pressure levels, and color coding (according to international standards).
- Observing how manifold systems are assembled, switched, and maintained.
- Learning about safety protocols including storage conditions, temperature control, and ventilation.
- Understanding how gas lines are traced and connected to wall-mounted terminal units in various hospital rooms.

This experience gave me a practical understanding of not only how essential gases are to hospitals, but also how biomedical engineers must regularly monitor, test, and ensure the safety of these pressurized systems.

My experience with the hospital's gas supply system was both exciting and enlightening. From witnessing large bulk storage tanks to understanding the inner workings of central supply rooms, I gained a deep appreciation for the engineering and safety measures involved in medical gas management. This aspect of the training greatly improved my biomedical engineering expertise, specifically my comprehension of how such systems support life-critical functions throughout the hospital.

## Conclusion

My two-month training at King Abdulaziz University Hospital was a wonderful and eye-opening experience that helped me gain a better grasp of biomedical engineering in a clinical setting. Throughout my rotation, I had the opportunity to visit and watch operations in a variety of departments, including Radiology, Laboratory, CSSD, ICU, Dialysis, OPD, and the Operating Room. Each area brought a distinct mix of medical equipment, technology, and technical difficulties, broadening my perspective beyond theoretical understanding. In departments such as CSSD and Radiology, I was able to directly witness equipment workflows, preventive maintenance methods, and the crucial importance of safety in infection control and radiation management. I also learned about the differences between diagnostic and therapeutic radiography, as well as how technologies like LINACs and CT scanners help with both diagnosis and therapy in advanced clinical care.

One of the most insightful and hands-on parts of the training was my experience with the hospital's medical gas supply system, where I explored the use, storage, and distribution of various gases such as oxygen, nitrous oxide, carbon dioxide, Entonox, helium, and specialized gas mixtures. I had the opportunity to inspect a wide range of gas cylinder sizes—from portable D and E cylinders to massive K-sized and bulk storage tanks—used to supply critical care units across the hospital. This exposure helped me understand how essential gas management systems are in sustaining life and supporting surgeries, emergency care, anesthesia, and respiratory therapy. It also highlighted the importance of biomedical engineering in maintaining pressure regulators, alarm panels, and safety protocols that ensure uninterrupted gas delivery.

One of the most significant lessons learned from this course was understanding how biomedical engineers contribute not just to equipment repair, but also to system safety, performance monitoring, and coordination with medical personnel. I was also aware of the constraints imposed by equipment warranties and clinical schedules, but these experiences taught me to adjust through observation, documentation review, and technical discussions. Overall, this educational program considerably improved my technical capabilities, professional awareness, and understanding of biomedical engineers' responsibilities in the hospital setting. I am convinced that the knowledge I obtained throughout this program will greatly benefit my future career in biomedical engineering.

# APPENDIX A

Appendix A – Comprehensive ICU Medical Devices Table (Extended)

| ICU Department | Device Name | Function / Description |
|---|---|---|
| SICU | Ventilator | Assists breathing post-surgery |
| | Infusion Pumps | Delivers medication/fluids continuously |
| | Patient Monitor | Displays ECG, BP, SpO, RR, Temperature |
| | Syringe Pump | Administers drugs with precise volume |
| | ECG Machine | Monitors heart activity post-op |
| | Defibrillator | Used for cardiac resuscitation |
| | Suction Unit | Clears airway secretions or fluids |
| | Anesthesia Monitor | Monitors anesthesia levels post-op |
| | Bedside Ultrasound | For emergency abdominal or vascular checks |
| | Surgical Lights | Focused illumination for post-op interventions |
| CCU | Cardiac Monitor | Continuous ECG and rhythm monitoring |
| | Defibrillator | Manages sudden cardiac arrest |
| | External Pacemaker | Provides temporary heart pacing |
| | Arterial BP Monitor | Invasive blood pressure monitoring |
| | Oxygen Concentrator | Provides high-flow oxygen |
| | Holter Monitor | 24–48 hour ECG tracking |

| | | |
|---|---|---|
| | Infusion Pump | Administers cardiac medications |
| | ECG Machine (12-lead) | Full cardiac assessment |
| | Pulse Oximeter | Measures blood oxygen saturation |
| MICU | Mechanical Ventilator | Critical respiratory support |
| | Hemodialysis Machine | Supports kidney failure patients |
| | Patient Monitor | Displays vital signs continuously |
| | Capnography Monitor | Measures end-tidal CO |
| | Bronchoscope | Lung evaluation for respiratory issues |
| | Blood Gas Analyzer | Assesses blood oxygenation and pH |
| | Suction Machine | Clears airways of fluids |
| | Nebulizer | Delivers inhaled medications |
| | Infusion & Syringe Pumps | Precise drug delivery |

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| ICU Depart-ment | Device Name | Function / Description |
|---|---|---|
| NICU | Incubator | Temperature/humidity controlled environment |
| | Infant Radiant Warmer | Maintains neonatal warmth |
| | Neonatal Ventilator | Respiratory support for preterm infants |
| | CPAP Machine (Infant) | Non-invasive respiratory support |
| | Phototherapy Unit | Treats neonatal jaundice |
| | Neonatal Monitor | HR, RR, SpO, Temperature tracking |
| | Syringe & Feeding Pumps | Milk/medication administration |
| | Infant Resuscitation Kit | Emergency newborn life support |
| | Transilluminator | Vein detection or air leak identification |

| | | |
|---|---|---|
| | Jaundice Meter | Non-invasive bilirubin testing |
| PICU | Pediatric Ventilator | Respiratory support for critically ill children |
| | Pediatric Monitor | Child-specific vital signs display |
| | Defibrillator (Pediatric) | Cardiac emergency treatment for children |
| | Infusion Pump | Medication/fluid administration |
| | Pediatric Suction Unit | Clears airway secretions |
| | Nebulizer | Delivers respiratory medications |
| | CPAP/BiPAP Machine | Non-invasive respiratory support |
| | Pediatric Oxygen Masks | Tailored oxygen therapy |
| | Pediatric Beds | Size-appropriate ICU beds with safety rails |
| PCICU | Pediatric Cardiac Monitor | Monitors arrhythmias, HR, BP, SpO |
| | ECMO Machine | Heart-lung bypass system for critical care |
| | External Pacemaker | Temporary post-op pacing |
| | Cardiac Output Monitor | Measures heart pumping efficiency |
| | Infusion/Syringe Pumps | Controlled drug administration |
| | Ventilator | Respiratory support for post-cardiac surgery |
| | Arterial Line Setup | Invasive BP and blood sampling |
| | Capnography Monitor | Measures CO levels |
| | Oxygen Blender | Delivers precise oxygen concentration |
| | Warming Mattress | Prevents hypothermia post-op |

## Figure 2 Appendix B (additional images)

**Appendix – Medical Device Images by Department**

**ICU Department**



Figure 1: ICU – Patient Monitor



Figure 2: ICU – Mechanical Ventilator

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